5. Which of the following is used to describe the worst case of an algorithm?

(A) Big A (B) Big S (C) Big W (D) Big O

6. Big Ω is the reverse of

(A) Big O (B) Big θ (C) Big A (D) Big S

7. Binary search is also called as

(A) Linear search (B) Sequential search

(C) Random search (D) Half-interval search

8. The Θ notation in asymptotic evaluation represents

(A) Base case (B) Average case (C) Worst case (D) NULL case

9. If a problem can be broken into subproblems which are reused several times, the problem possesses which property?

(A) Overlapping subproblems (B) Optimal substructure

(C) Memoization (D) Greedy

10. In dynamic programming, the technique of storing the previously calculated values is called ?

(A) Saving value property (B) Storing value property

(C) Memoization (D) Mapping

Part - II

Answer the following questions **(2 Marks)**

1. What is an Algorithm?
2. Write the phases of performance evaluation of an algorithm.
3. What is Insertion sort?
4. What is Sorting?
5. What is searching? Write its types.

Part - III

Answer the following questions **(3 Marks)**

1. List the characteristics of an algorithm.
2. Discuss about Algorithmic complexity and its types.

12th Computer Science_EM Chapter 4.indd 44 21-12-2022 15:55:01